

Fiscal Policy

ECON 101 – Principles of Macroeconomics

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- **12.1 What Is Fiscal Policy?**
- 12.2 The Effects of Fiscal Policy on Real GDP and the Price Level
- 12.4 Government Purchases and Tax Multipliers
- 12.5 The Limits of Fiscal Policy as a Stimulus
- 12.6 Deficits, Surpluses, and Federal Government Debt
- 12.7 The Effects of Fiscal Policy in the Long Run
- **Appendix A Closer Look at the Multiplier**

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12.1 Learning Objective

Goal

Define fiscal policy and distinguish between automatic stabilizers and discretionary fiscal policy.

- **Fiscal policy:** changes in **federal government taxes and spending** intended to achieve macroeconomic objectives.
 - Provincial spending is important but primarily aimed at service delivery, not national macrostabilization.
- The federal government can **shift aggregate demand** by changing spending and tax decisions.
- **Canada 2020 example:** the federal government launched \$314 billion in emergency spending (CERB, CEWS, CERS) — the largest fiscal intervention in Canadian history.

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Automatic Stabilizers vs. Discretionary Fiscal Policy

Automatic Stabilizers

Spending and taxes that automatically change with the business cycle **without new legislation**.

- Employment Insurance (EI): rises in recessions as more workers qualify.
- Income tax: falls in recessions (lower incomes $\Rightarrow$ lower tax revenue).
- Social assistance: rises when unemployment increases.

Act fast, no legislative delay.

Discretionary Fiscal Policy

Intentional government actions requiring new legislation.

- 2009 Economic Action Plan: \$40B infrastructure stimulus.
- 2020 CERB: \$2,000/month to unemployed Canadians.
- 2024 GST holiday: temporary sales tax cut.
- 2025 income tax cut proposal.

Takes time; subject to political process.

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Goal

Explain how fiscal policy affects aggregate demand and how it can stabilize the economy.

- **Two channels:**
 - Changes in government purchases (G): affects AD **directly**.
 - Changes in taxes (T): affects disposable income $\Rightarrow$ consumption $\Rightarrow$ AD **indirectly**.
- **Expansionary fiscal policy:** $\uparrow G$ or $\downarrow T \Rightarrow$ AD shifts right.
- **Contractionary fiscal policy:** $\downarrow G$ or $\uparrow T \Rightarrow$ AD shifts left.

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Figure 12.1 – Expansionary Fiscal Policy (Closing a Recessionary Gap)

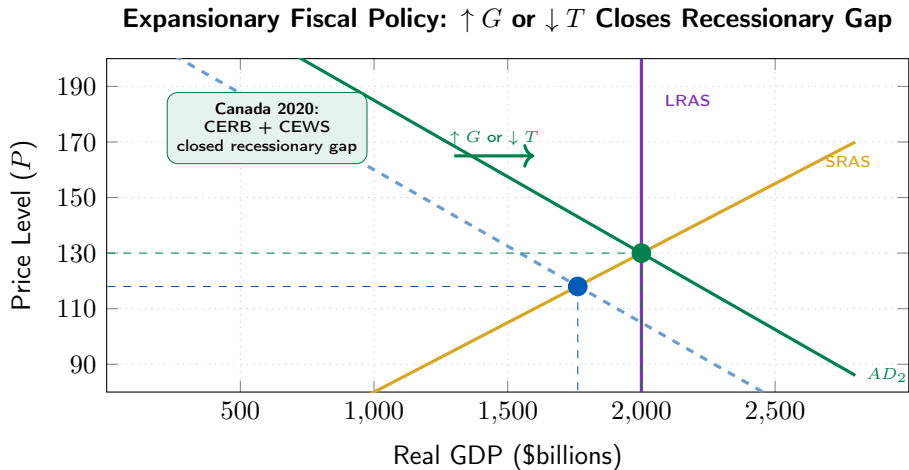


Figure 12.2 – Contractionary Fiscal Policy (Fighting Inflation)

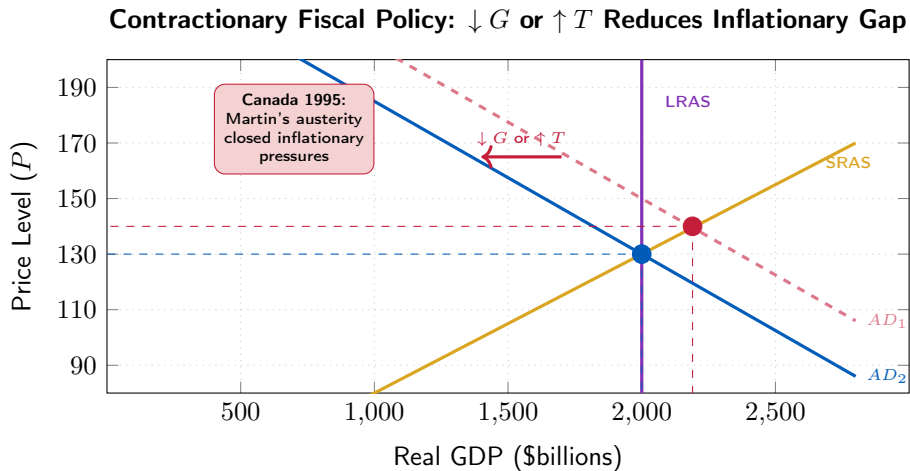


Table 12.1 – Countercyclical Fiscal Policy in Canada

Problem	Policy Type	Government Action	Canadian Example
Recession	Expansionary	$\uparrow G$ or $\downarrow T \Rightarrow$ AD shifts right	CERB/CEWS (\$314B) 2020
Rising inflation	Contractionary	$\downarrow G$ or $\uparrow T \Rightarrow$ AD shifts left	1995 Martin budget cuts

- **Note:** in Table 12.1 the original text references the U.S. Congress. The Canadian equivalent is Parliament — specifically the House of Commons, which must pass all spending bills and tax changes.
- Countercyclical policy aims to **dampen** the business cycle, not eliminate it.

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Explain how the government purchases and tax multipliers work.

- A \$1 increase in government spending raises equilibrium GDP by **more than \$1**.
- Why?
 - Initial spending $\Rightarrow$ incomes rise $\Rightarrow$ consumption rises $\Rightarrow$ incomes rise further $\Rightarrow$...
 - This chain is the **multiplier effect**.
- Government purchases multiplier:

$$\frac{\Delta Y}{\Delta G} = \frac{1}{1 - \text{MPC}}.$$

- Tax multiplier:

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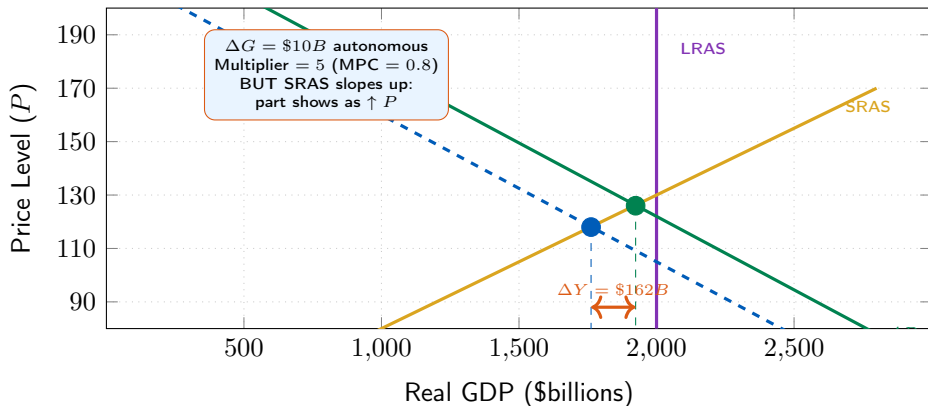
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Figure 12.3 – The Multiplier Effect in AD-AS

Multiplier: \$10B Increase in G Raises Equilibrium GDP by \$50B ($MPC = 0.8$)



Government Purchases Multiplier vs. Tax Multiplier

Government Purchases Multiplier

$$\frac{\Delta Y}{\Delta G} = \frac{1}{1 - \text{MPC}}$$

- \$1 of G **directly** adds \$1 to AD.
- Then the induced consumption chain kicks in.
- With $\text{MPC} = 0.8$: multiplier = 5.
- $\Delta G = \$10B \Rightarrow \Delta Y = \$50B$.

Tax Multiplier

$$\frac{\Delta Y}{\Delta T} = -\frac{\text{MPC}}{1 - \text{MPC}}$$

- Tax cut raises disposable income.
 - Households **save MPS** portion, spend MPC portion.
 - First-round effect: $\$ \text{MPC}$ (not \$1).
 - With $\text{MPC} = 0.8$: tax multiplier = -4 .
 - $\Delta T = -\$10B \Rightarrow \Delta Y = +\$40B$.
-
- The **government purchases multiplier is larger** (absolute value) than the tax multiplier: spending injection is fully spent, while a tax cut is partly saved.

12.5 Learning Objective

Goal

Discuss the difficulties that can arise in implementing fiscal policy.

- Fiscal policy is powerful in theory but faces practical limitations:
 1. **Timing problems** (legislative and implementation lags).
 2. **Crowding out** (higher government borrowing raises interest rates, reducing private spending).
 3. **Political constraints** (fiscal policy is a political process, not just an economic one).

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Timing Problems

1. Recognition lag:

- GDP data arrives 2–3 months after the quarter ends. By the time a recession is recognized, it may already be deep.

2. Legislative delay:

- Parliament must debate and approve spending or tax changes.
- **Canadian example (2009):** Conservative minority government's Economic Action Plan took months to implement after the recession had begun.

3. Implementation delay:

- Infrastructure projects take months to years to actually begin (design, tender, construction).
- **"Shovel-ready" projects:** a political priority in 2009 precisely because of this lag.

4. Because of these delays, fiscal stimulus may arrive **after the recession has ended** — adding fuel to the next boom.

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A decline in private expenditures (C, I, NX) that results from an increase in government purchases.

- **Mechanism:**

- Government increases spending $\Rightarrow$ borrows more (issues more bonds).
- Higher bond supply $\Rightarrow$ bond prices fall $\Rightarrow$ interest rates rise.
- Higher interest rates $\Rightarrow$ lower private investment, lower consumer credit, stronger CAD (lower NX).
- Private spending falls — partially offsetting the stimulus.

- **Canadian context:**

- Canada's federal debt is $\approx$ \$1.3 trillion (2024).
- Higher debt-to-GDP ratios may gradually raise borrowing costs.
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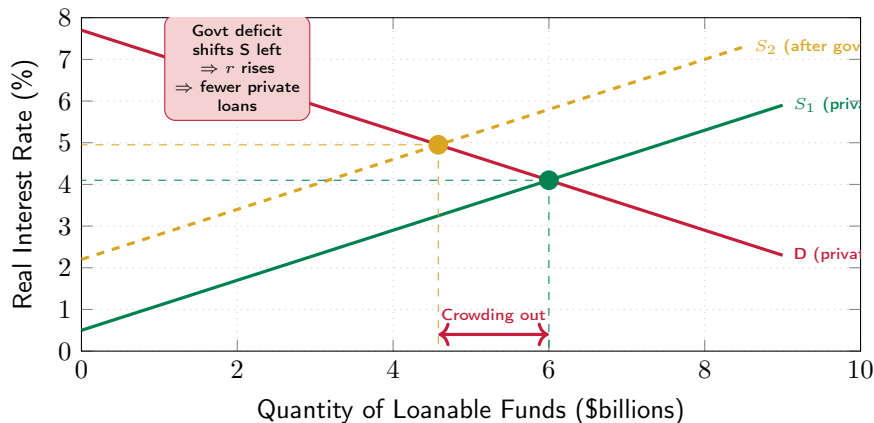
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Figure 12.4 – Crowding Out in the Loanable Funds Market

Government Borrowing Raises Interest Rates, Crowding Out Private Investment



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Define budget deficit and federal government debt, and explain how the budget serves as an automatic stabilizer.

- **Budget deficit:** expenditures $>$ tax revenues.
- Budget surplus: expenditures $<$ tax revenues.
- Federal government debt (national debt): the total stock of outstanding Government of Canada bonds.
- Canadian fiscal history:
 - Deficits almost every year since 1970.
 - One stretch of surpluses: 1998–2008 (Chrétien/Martin era).
 - 2020–21: deficit of \$314 billion — by far the largest ever.
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 - 2024–25: deficit $\approx$ \$40 billion.

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Define budget deficit and federal government debt, and explain how the budget serves as an automatic stabilizer.

- **Budget deficit:** expenditures $>$ tax revenues.
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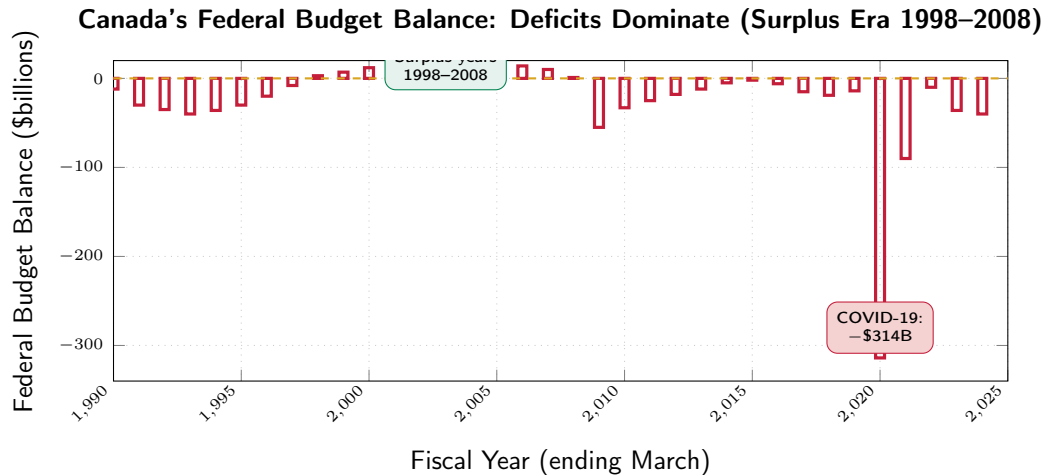
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Figure 12.5 – Canada's Federal Budget Balance (1990–2024)



Source: Department of Finance Canada, Federal Budget. Positive = surplus; negative = deficit.

Automatic Stabilizers and the Business Cycle

- **In recessions** (automatic stabilizers kick in):
 - Tax revenues **fall** (lower incomes, profits, consumption).
 - EI payments, social assistance **rise**.
 - Result: budget deficit grows automatically, helping support AD.
- **In expansions:**
 - Tax revenues **rise**.
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 - Result: budget moves toward surplus automatically, dampening inflation.
- **Canada's EI example (2020):**
 - EI and emergency benefits paid out $\approx$ \$100 billion in 2020 alone.
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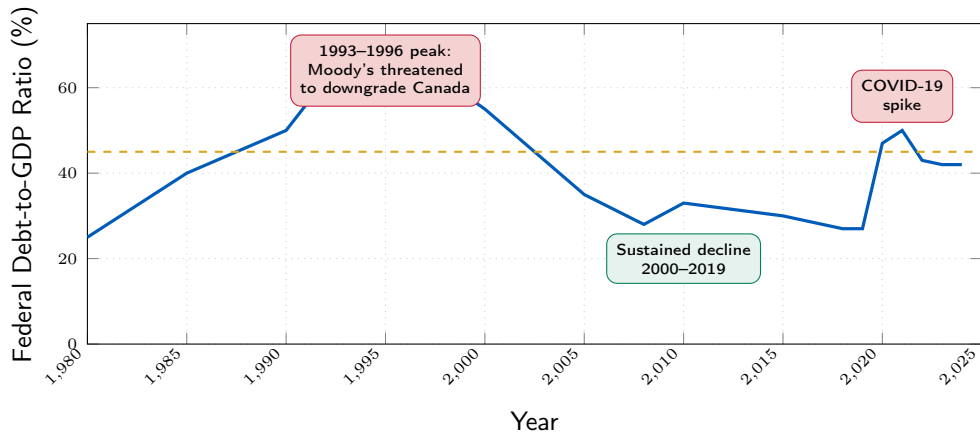
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Figure 12.6 – Federal Government Debt to GDP Ratio (Canada)

Federal Debt-to-GDP Rose Sharply in 1990s, Then Fell, Then Rose Again (2020)



Source: Department of Finance Canada; IMF Fiscal Monitor. G7 average is approximate.

Is Government Debt a Problem?

- **Short-term view:** Canada's federal debt-to-GDP ratio ($\approx 42\%$) is lower than most G7 countries. Not an immediate crisis.
- **Long-term concerns:**
 - If debt grows faster than GDP, interest payments consume a larger share of the budget.
 - **Crowding out:** rising debt may gradually push up real interest rates, reducing private investment.
 - **Generational equity:** future taxpayers (today's children) inherit the debt.
- **When is debt less problematic?**
 - When it finances **productive investments**: infrastructure, education, R&D.
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Discuss the effects of fiscal policy in the long run (supply-side effects).

- So far: fiscal policy as a short-run stabilization tool (shifts AD).
- Now: how fiscal policy affects **potential GDP** (shifts LRAS).
- **Supply-side economics:**
 - Fiscal policy actions aimed at increasing long-run aggregate supply.
 - Focus: taxes and regulations that affect incentives to work, save, invest, and innovate.

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The Tax Wedge and Incentives

- Most taxes create a **wedge** between the pre-tax and post-tax return to an activity.
- Examples:
 - **Personal income tax:** if your marginal rate is 53% (federal + BC combined), each extra dollar earned yields only 47 cents after tax.
 - High rates may reduce hours worked, entrepreneurship, or human capital investment.
 - **Corporate income tax:** Canada's combined rate $\approx 26\%$ in 2024.
 - High rates may reduce investment in new machinery, R&D, and job creation.
 - **Capital gains tax:** affects saving and investment decisions.
 - In 2014, Canada's Trudeau government raised capital gains inclusion rate from 50% to 67% → reduced the after-tax return on investment in real estate.

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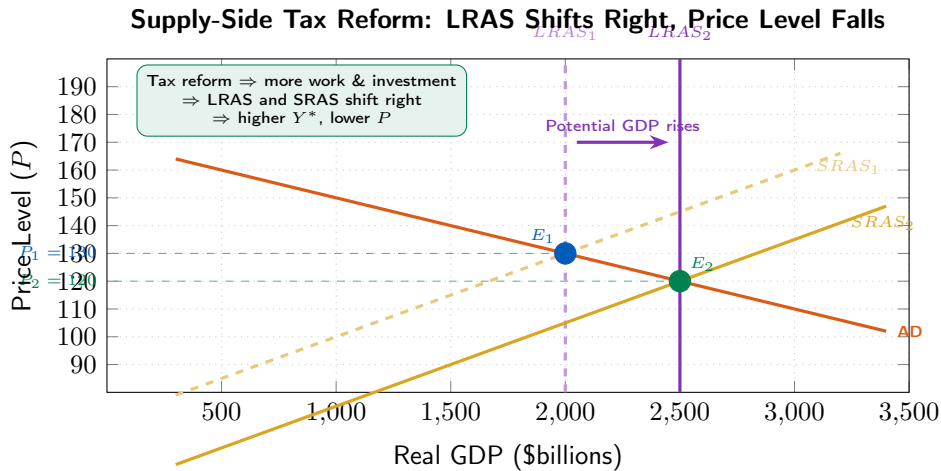
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Figure 12.7 – Supply-Side Effects of a Tax Cut



Supply-side tax reform raises potential GDP from \$2,000B to \$2,500B while lowering the long-run price level.

Long-Run Effects of Tax Reform: What We Know

- Tax reform can increase potential GDP in the long run by:
 - Raising labour supply (more hours worked, more participation).
 - Increasing saving and investment (more capital formation).
 - Encouraging entrepreneurship and innovation.
- **Uncertainty:**
 - Magnitude of supply-side effects is **empirically contested**.
 - People may want to work more at lower marginal rates, but may face *constraints* (fixed schedules, childcare).
 - Tax cuts also create short-run demand effects (shifting AD right), which may dominate.
- **Canadian policy debate:**
 - 2024 capital gains inclusion rate increase: critics argued it would deter investment; supporters argued revenue would fund social priorities.
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Appendix – The Government Purchases and Tax Multipliers

- Simple Canadian macroeconomic model (closed economy, fixed price level):

$$\begin{aligned}C &= 1,000 + 0.8(Y - T), & T &= 250, \\I &= 300, & G &= 500.\end{aligned}$$

- Equilibrium: $Y = C + I + G$:

$$Y = 1,000 + 0.8(Y - 250) + 300 + 500.$$

$$Y = 1,600 + 0.8Y - 200 = 1,400 + 0.8Y.$$

$$0.2Y = 1,400 \quad \Rightarrow \quad Y = \$7,000 \text{ billion}.$$

- Government purchases multiplier: $\frac{\Delta Y}{\Delta G} = \frac{1}{1-0.8} = 5$.
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The Tax Multiplier and the Balanced Budget Multiplier

- **Tax multiplier:**

$$\frac{\Delta Y}{\Delta T} = -\frac{\text{MPC}}{1 - \text{MPC}} = -\frac{0.8}{0.2} = -4.$$

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- **Why smaller than G multiplier?** A \$10B tax cut gives \$10B to households, but they save $0.2 \times 10 = \$2B$, so only \$8B is initially spent.
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